Data Science – Bivariate Analysis

1)Replace the NaN values with correct value. And justify why you have chosen the same.

```
sl_no             0
gender            0
ssc_p             0
ssc_b             0
hsc_p             0
hsc_b             0
hsc_s             0
degree_p          0
degree_t          0
workex            0
etest_p           0
specialisation    0
mba_p             0
status            0
salary           67
dtype: int64
```

The `salary` column contained missing values corresponding to students who were not placed. Since these students did not receive any salary package, the missing values were replaced with **0**. This approach preserves the meaning of the data and avoids introducing misleading salary values that would result from using the mean or median.

2)How many of them are not placed?

```
Not Placed    67
```

3)Find the reason for non placement from the dataset?

	ssc_p	hsc_p	degree_p	etest_p	mba_p
status					
Not Placed	57.544030	58.395522	61.134179	69.587910	61.612836
Placed	71.721486	69.926554	68.740541	73.238041	62.579392

The data suggests that **lower SSC, HSC, and Degree percentages are strongly associated with not being placed**.

4)What kind of relation between salary and mba_p

	salary	mba_p
salary	1.000000	0.175013
mba_p	0.175013	1.000000

There is a weak positive correlation (**r = 0.175**) between MBA percentage and salary. Students with higher MBA percentages tend to receive slightly higher salaries, but the association is not strong. Other factors likely have a greater influence on salary than MBA percentage alone.

5)Which specialization is getting minimum salary?

```
specialisation
Mkt&Fin     200000.0
Mkt&HR      200000.0
Name: salary, dtype: float64
```

Both Marketing & Finance (Mkt&Fin) and Marketing & HR (Mkt&HR) have the same minimum salary of ₹200,000**.**

6)How many of them getting above 500000 salary?

```
specialisation  sl_no  gender  ssc_p  ssc_b    hsc_p  hsc_b    hsc_s     degree_p  degree_t   workex  etest_p  mba_p  status  salary
Mkt&Fin         120    M       60.8   Central  68.40  Central  Commerce  64.6      Comm&Mgmt  Yes     82.66    64.34  Placed  940000.0    1
                151    M       71.0   Central  58.66  Central  Science   58.0      Sci&Tech   Yes     56.00    61.30  Placed  690000.0    1
                178    F       73.0   Central  97.00  Others   Commerce  79.0      Comm&Mgmt  Yes     89.00    70.81  Placed  650000.0    1
Name: count, dtype: int64
```

Three students received salaries above ₹500,000, and all of them were from the Marketing & Finance specialization. The highest salary among them was ₹940,000.

7)Test the Analysis of Variance between etest_p and mba_p at signifance level 5%.(Make decision using Hypothesis Testing)

```
F_onewayResult(statistic=np.float64(98.64487057324708), pvalue=np.float64(4.672547689
133693e-21))
```

At the 5% significance level, there is sufficient evidence to conclude that the mean `etest_p` score differs significantly from the mean `mba_p` score. Therefore, a statistically significant difference exists between `etest_p` and `mba_p`

8)Test the similarity between the degree_t(Sci&Tech) and specialisation(Mkt&HR) with respect to salary at significance level of 5%.(Make decision using Hypothesis Testing)

At the 5% significance level, there is sufficient evidence to conclude that the mean salary of `Sci&Tech` degree holders differs significantly from the mean salary of `Mkt&HR` specialization students.

9)Convert the normal distribution to standard normal distribution for salary column

	salary	salary_zscore
0	270000.0	-0.200292
1	200000.0	-0.951839
2	250000.0	-0.415019
4	425000.0	1.463849
7	252000.0	-0.393547

- The salary of **₹425,000** is **above the average salary**, as its Z-score is **+1.464**.
- The salaries **₹200,000, ₹250,000, ₹252,000, and ₹270,000** are **below the average salary**, since their Z-scores are negative.
- After standardization, the transformed variable (`salary_zscore`) has:
  - Mean ≈ 0
  - Standard Deviation ≈ 1

10)What is the probability Density Function of the salary range from 700000 to 900000?

```
Mean: 288655.4054054054
Expand Output 241958875
z1: 4.401410309664898
z2: 6.541421564220343
Porbability: 5.377445759013888e-06
```

The probability that a student's salary lies between **₹700,000 and ₹900,000** is approximately **0.00000538** (or **0.00054%**). This is an extremely small probability, indicating that salaries in this range are very rare in the dataset.

11)Test the similarity between the degree_t(Sci&Tech)with respect to etest_p and mba_p at significance level of 5%.(Make decision using Hypothesis Testing)

`etest_p` and `mba_p` are **not similar** for Sci&Tech students at the 5% significance level

12)Which parameter is highly correlated with salary?

`etest_p` (Employability Test Percentage) has the highest correlation with salary, with a correlation coefficient of **0.1783**.

13) plot any useful graph and explain it.

The graph indicates that **Marketing & Finance students receive higher average salaries than Marketing & HR students**. Therefore, among the two specializations, **Mkt&Fin appears to offer better salary outcomes** in this dataset. This suggests that specialization may have an influence on salary packages received during placements.